PROGRAM CODE :

```
clc;
clear;
k=10;
num=poly([1],"s","coeff");
den=poly([0 1 0.6 0.05],"s","coeff");
s1=syslin('c',num,den);
G=k*s1;
disp(G, 'The given transfer function G(s)=');
bode(G,0.01,1000);
xtitle('Bode plot of the given transfer function G(s)');
[g,frp]=g_margin(G);
[p,frg]=p_margin(G)
show_margins(G)
disp(frg,'Gain crossover frequency=',p,'Phase margin(degrees)=');
disp(frp,'Phase crossover frequency=',g,'Gain margin(dB)='); clc;
clear;
k=10;
num=poly([1],"s","coeff");
den=poly([0 1 0.6 0.05],"s","coeff");
s1=syslin('c',num,den);
G=k*s1;
disp(G, 'The given transfer function G(s)=');
bode(G,0.01,1000);
xtitle('Bode plot of the given transfer function G(s)');
[g,frp]=g_margin(G);
[p,frg]=p_margin(G)
show_margins(G)
```

```
disp(frg,'Gain crossover frequency=',p,'Phase margin(degrees)=');

disp(frp,'Phase crossover frequency=',g,'Gain margin(dB)=');
```

OUTPUT :

